III a) Match Column I with Columns II and III

Column I	Column II	Column III
Auxin	*Gibberella fujikuroi*	Abscission
Ethylene	Coconut milk	Internodal elongation
Abscisic acid	Coleoptile tip	Apical dominance
Cytokinin	Chloroplast	Ripening
Gibberellins	Fruits	Cell division

III b) Match the following hormones with their deficiency states

Hormones		*Disorders*
a) Thyroxine	-	Acromegaly
b) Insulin	-	Tetany
c) Parathormone	-	Simple goitre
d) Growth hormone	-	Diabetes insipidus
e) ADH	-	Diabetes mellitus

IV State whether True or false, If false write the correct statement

1. A plant hormone concerned with stimulation of cell division and promotion of nutrient mobilization is cytokinin.
2. Gibberellins cause parthenocarpy in tomato.
3. Ethylene retards senescence of leaves, flowers and fruits.
4. Exopthalmic goiter is due to the over secretion of thyroxine.
5. Pituitary gland is divided into four lobes.
6. Estrogen is secreted by corpus luteum.

V Assertion and Reasoning

Direction: In each of the following questions a statement of assertion (A) is given and a corresponding statement of reason (R) is given just below it. Mark the correct statement as.

a. If both A and R are true and R is correct explanation of A

b. If both A and R are true but R is not the correct explanation of A

c. A is true but R is false

d. Both A and R are false

1. **Assertion:** Application of cytokinin to marketed vegetables can keep them fresh for several days.

   Reason: Cytokinins delay senescence of leaves and other organs by mobilisation of nutrients.

2. **Assertion (A):** Pituitary gland is referred as "Master gland".

   Reason (R): It controls the functioning of other endocrine glands.

3. **Assertion (A):** Diabetes mellitus increases the blood sugar levels.

   Reason (R): Insulin decreases the blood sugar levels.

VI Answer in a word or sentence

1. Which hormone promotes the production of male flowers in Cucurbits?
2. Write the name of a synthetic auxin.
3. Which hormone induces parthenocarpy in tomatoes?
4. What is the hormone responsible for the secretion of milk in female after child birth?
5. Name the hormones which regulates water and mineral metabolism in man.
6. Which hormone is secreted during emergency situation in man?
7. Which gland secretes digestive enzymes and hormones?
8. Name the endocrine glands associated with kidneys.

VII Short answer questions

1. What are synthetic auxins? Give examples.
2. What is bolting? How can it be induced artificially?

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